RoPE Is a Substrate, Not a Trick: Structural Separation between Toroidal Composition and Transformer Primitives on Finite Groups

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Abstract

We study the representation of finite-group actions by the four composition primitives that appear in contemporary transformer architectures. For any finite group G with a non-trivial element of order $m \geq 2$, additive residual updates on a Euclidean embedding $\mathbb{R}^d$ cannot realise any non-trivial G-action at any depth, width, or training budget (Theorem 3); the argument uses only the torsion-freeness of $\mathbb{R}^d$. We then prove a four-way *complexity separation* (Theorems 7, 8, 9, 10): (P0) the additive primitive is broken from $K=1$; (P1) the rotary positional layer (RoPE) is structurally isomorphic to the $\mathbb{T}^1$ substrate restricted to one phase, and achieves drift $O(K \cdot \varepsilon_{\text{mach}})$; (P2) an MLP primitive with generic non-linearity carries a strictly positive per-step approximation residual $\tilde{\varepsilon} > 0$, yielding drift $\Omega(K \cdot \tilde{\varepsilon})$ and a decoder collapse at composition depth $K^* \approx \pi/(n \cdot \tilde{\varepsilon})$; (P3) the $\mathbb{T}^N$ substrate has drift $O(K \cdot \varepsilon_{\text{mach}})$ and stays at machine precision up to $K \approx 10^{15}$. Empirically, on iterated $\mathbb{Z}_n$ composition at $n \in \{100, 1000, 360\}$ and composition depth K up to 10^6, (P2) collapses at precisely the predicted K^*, (P0) is broken throughout, and (P1, P3) stay at 100%. A real small instruction-tuned LLM with chain-of-thought (Qwen2.5-0.5B) constitutes a fourth, *closed-form* arithmetic regime: on $(r+K) \bmod n$ it is noisy (58% mean accuracy) and *not* monotonically worse with K because it exploits algebraic shortcuts, sharpening the claim that Thm 9 applies to iterated-primitive compositions, not to every neural arithmetic. On a state-dependent walk $p_{k+1} = (p_k + \Delta[\rho(p_k)]) \bmod n$ that admits no closed form, the same LLM scores 0/60 across $K \in \{1, 2, 5, 10, 20, 50\}$ with a diagnostic failure mode of ignoring the in-prompt role table, while the substrate scores 60/60 at $\mathcal{O}(d)$ per step. Finally, we show that the substrate is not engineered: gradient descent on the cosine contrastive loss recovers the oracle homomorphism up to the natural symmetry $\mathrm{Aut}(\mathbb{Z}_n) \times S^1 \times \mathbb{Z}_2$ (Theorems 12–16), and compositional closure $n\rho_k \equiv 0 \pmod{2\pi}$ emerges from the training data at machine precision $< 10^{-14}$. The rotary positional layer's empirical success on compositional tasks is thus *a measurement of the* $\mathbb{T}^1$ *substrate*, not an engineering coincidence; and the $\mathbb{T}^N$ substrate is the unique geometric object that gradient descent finds when asked to compose.

1 Introduction

Transformers famously fail on modular arithmetic and parity—tasks that an elementary-school student solves by counting [2, 7, 11]—yet succeed, sometimes dramatically, on *some* compositional tasks [4, 8, 9]. What discriminates the successes from the failures?

We identify the discriminator as a *geometric* property of the architecture, not a scale or prompting one. Every finite-group action $g \mapsto g^K \cdot x$ admits a closed-form K-th power on a *toroidal* embedding $\mathbb{T}^N$ via phase addition modulo 2π; no torsion-free Euclidean embedding $\mathbb{R}^d$ can realise any non-trivial finite-group action at any d (§3, Thm 3). The rotary positional layer (RoPE) is

a 2×2 rotation acting on a pair of embedding coordinates; on a circular token embedding it is an *isometry* onto a subcircle of $\mathbb{T}^1$ and realises $\mathbb{Z}_n$ exactly (§4, Thm 8). An MLP approximating the same rotation by a trained non-linear map carries a strictly positive per-step approximation residual $\tilde{\varepsilon}$, and its K-fold iterate collapses at $K^* \approx \pi/(n \cdot \tilde{\varepsilon})$ (Thm 9). The $\mathbb{T}^N$ substrate is a drop-in algebraic primitive that stays at machine precision for K up to 10^{15} (Thm 10).

The picture is sharp but needs one honest qualification. A real instruction-tuned LLM with chain-of-thought does not iterate a primitive K times; it attempts a closed-form modular reduction. In empirical tests the LLM baseline is catastrophically wrong on trivial cases (20% at $K = 1$) and nearly correct on apparently hard cases where the operands admit an algebraic shortcut (80–90% at $K = 10\,000$). The separation in Theorems 7–10 therefore describes *iterated primitive classes*, not LLM arithmetic in general. We include the real LLM baseline in §5 to make this distinction explicit: the substrate's advantage is *uniformity of correctness*, not the monotonic-collapse story a naive reading of Thm 9 would suggest.

Contributions.

C1. A universal structural impossibility result (Thm 3): no additive Euclidean representation realises any non-trivial finite group.

C2. A four-primitive complexity separation (Theorems 7–10) quantified in K-fold composition drift.

C3. Identification of RoPE as a $\mathbb{T}^1$ substrate (Thm 8): its empirical success on compositional tasks is a measurement of the substrate's capacity, not an engineering trick.

C4. Learnability: gradient descent recovers the oracle homomorphism up to $\mathrm{Aut}(\mathbb{Z}_n) \times S^1 \times \mathbb{Z}_2$ symmetry (Theorems 12–16), with compositional closure emerging from data at machine precision.

We connect to grokking [4, 8, 9], Lie-RoPE extensions [10, 16], and transformer complexity bounds [1, 7, 11] in §8.

2 Background and setup

The finite-group composition task. Fix a finite group G, a generator $g \in G$, and $x_0 \in G$. The K-fold composition task is: given K, produce the decoded state $g^K \cdot x_0$. Benchmark groups in this paper are $G = \mathbb{Z}_n$ with $n \in \{7, 12, 100, 360, 1000\}$, and the dihedral groups D_n.

Embedding and decoder. An embedding $\rho : G \to M$, $M \subseteq \mathbb{R}^d$ or $M \subseteq \mathbb{T}^N$, assigns a point $\rho(g) \in M$ to every group element. The decoder $\mathrm{dec} : M \to G$ is nearest-neighbour in the ambient metric (ℓ^2 on $\mathbb{R}^d$, circular distance on $\mathbb{T}^N$), with resolution half-gap $\varepsilon_{\min}/2 := \pi/n$ on $\mathbb{T}^1$. A decoding is *correct* if $\mathrm{dec}(\hat{x}) = x_{\mathrm{true}}$.

Four primitives. A composition primitive P is a map $P : M \to M$ that is applied per layer. We consider four classes:

(P0) Additive residual.
$P(x) = x + r$, $r \in \mathbb{R}^d$ fixed.

(P1) Linear / RoPE.
$P(x) = Wx$, $W \in \mathbb{R}^{d \times d}$; the RoPE case is $W = R(2\pi/n) \oplus I$ acting on a 2-plane.

(P2) MLP / FFN.
$P(x) = \sigma(W_2\sigma(W_1x + b_1) + b_2)$.

(P3) $\mathbb{T}^N$ substrate.
$\phi \mapsto \phi + \rho(g)$ on $\mathbb{T}^N$.

Drift. For $K \geq 1$, the K-fold drift is $\Delta_K := \|P^K(x_0) - x_0^{\text{true}}\|$ in the appropriate metric. We measure three regimes: machine precision $\varepsilon_{\text{mach}} \approx 2^{-52}$ (IEEE-754 double), per-step approximation residual $\tilde{\varepsilon} > 0$ for P2, and decoder half-gap $\varepsilon_{\min}/2 = \pi/n$.

3 Structural impossibility of $\mathbb{R}^d$ additive representations

The argument is short but surprisingly strong.

Theorem 1 ($\mathbb{Z}_n$ impossibility)**.** *Let $n \geq 2$ and let $\rho : \mathbb{Z}_n \to \mathbb{R}^d$ be additive: $\rho(a + b) = \rho(a) + \rho(b)$. Then $\rho \equiv 0$.*

Proof. For any generator $g \in \mathbb{Z}_n$ of order m, additivity gives $m\rho(g) = 0$ in $\mathbb{R}^d$. Since $\mathbb{R}^d$ is torsion-free, $\rho(g) = 0$. □

Corollary 2 (Finite abelian products)**.** *For any finite abelian $G = \prod_i \mathbb{Z}_{n_i}$, the only additive representation $\rho : G \to \mathbb{R}^d$ is trivial.*

Theorem 3 (Universal finite-group impossibility)**.** *Let G be any finite group with a non-trivial element of order $m \geq 2$, and let $\rho : G \to \mathbb{R}^d$ satisfy $\rho(gh) = \rho(g) + \rho(h)$. Then $\rho \equiv 0$.*

Proof. Thm 1 applied to the cyclic subgroup $\langle g \rangle$. □

The construction side recovers every finite abelian group on a torus and every finite group via matrix representation.

Theorem 4 ($\mathbb{Z}_n$ on $\mathbb{T}^1$; constructive)**.** *The map $\rho : \mathbb{Z}_n \to \mathbb{T}^1$, $\rho(k) = 2\pi k/n$, realises $\mathbb{Z}_n$ exactly, with decoder half-gap π/n and composition-depth-independent correctness.*

Theorem 5 (Finite abelian on $\mathbb{T}^k$)**.** *Every finite abelian group $G = \prod_i \mathbb{Z}_{n_i}$ admits an exact representation on $\mathbb{T}^k$ with $k = \#\{i : n_i \geq 2\}$.*

Theorem 6 (Non-abelian via matrix rep)**.** *Every finite group G admits a faithful orthogonal representation $\rho : G \to \mathrm{O}(d)$; for $G = D_n$, $d = 2$ suffices via $\rho(r) = R(2\pi/n)$, $\rho(s) = \mathrm{diag}(1, -1)$.*

Detailed proofs are in Appendix A.

4 Complexity separation on transformer primitives

We now bound the four primitives of §2 on the iterated $\mathbb{Z}_n$ task. The four theorems below are the paper's headline.

Theorem 7 (P0 additive — impossibility)**.** *For any G with an element of order $m \geq 2$ and any primitive P0 $x \mapsto x + r$ with $r \in \mathbb{R}^d$, the decoded accuracy on the task $x \mapsto g^K \cdot x$ is at most $1/|G| + o(1)$ as $|G| \to \infty$ for all $K \geq 1$ and all d.*

Proof. The K-fold application of P0 gives $x + Kr$. If this realises g^K at $K = m$ then $mr = 0$ in $\mathbb{R}^d$, forcing $r = 0$ (Thm 3). □

Theorem 8 (P1 RoPE is a $\mathbb{T}^1$ substrate)**.** *Let $W = R(2\pi/n) \in \mathrm{SO}(2)$ act on the circular embedding $x_k = (\cos 2\pi k/n, \sin 2\pi k/n)$. Then for every K, $\|W^K x_k - x_{(k+K) \bmod n}\| \le C \cdot K \cdot \varepsilon_{\mathrm{mach}}$ in floating-point arithmetic, and the map $R(\theta) \leftrightarrow e^{i\theta} \in U(1) \subset \mathbb{T}^1$ is an isometry.*

Remark (Why this is the paper's bomb). RoPE's empirical success on modular-arithmetic and position-tracking benchmarks is a *measurement* of $\mathbb{T}^1$ capacity. Every transformer that succeeds on a $\mathbb{Z}_n$ compositional task contains, at each rotary channel, a $\mathbb{T}^1$-substrate isomorph; every transformer that lacks rotary (or equivalent unitary) structure falls under Thm 7.

Theorem 9 (P2 MLP has strictly positive per-step residual)**.** *Let $f_\theta : \mathbb{R}^d \to \mathbb{R}^d$ be a bounded-width MLP with generic non-linearity (tanh, ReLU, GELU) trained on the single-step task $x_k \mapsto x_{k+1}$ on a $\mathbb{Z}_n$ circle. Then there exists $\tilde{\varepsilon} > 0$, depending on width, data, and optimiser but not on K, such that $\mathbb{E}\|f_\theta(x) - target\| \ge \tilde{\varepsilon}$. Hence $\|f_\theta^K(x_0) - x_{0+K}\| = \Omega(K \cdot \tilde{\varepsilon})$ and decoding accuracy drops below random-chance when $K \cdot \tilde{\varepsilon}$ exceeds the half-gap $\varepsilon_{\min}/2 = \pi/n$, i.e. at*

$$K^* \approx \frac{\pi}{n \cdot \tilde{\varepsilon}}.$$

Theorem 10 (P3 substrate drift is machine precision)**.** *The $\mathbb{T}^N$ substrate with exact phase addition has drift $\Delta_K \le K\epsilon_{\mathrm{mach}} 2\pi$ per coordinate. Decoding accuracy is 100% for all K with $K\epsilon_{\mathrm{mach}} N < \varepsilon_{\min}$; for $n \le 10^6$, K up to 10^{15}.*

Corollary 11 (The complexity separation, table form)**.** *For the iterated $\mathbb{Z}_n$ generator task at depth K:*

P0 broken from $K{=}1$; *P2 broken at $K{\approx}\pi/(n{\cdot}\tilde{\varepsilon})$;* *P1 and P3 unbroken throughout, with $P1 \subseteq P3$.*

Detailed proofs and error analyses are in Appendix B.

5 Empirical validation

We validate Thms 7–10 in three progressively more realistic regimes.

5.1 Primitive-class benchmark on $\mathbb{Z}_{12}$

Four primitives (P0, P1, P2, P3) were trained or defined on the single step $x_k \to x_{k+1}$ of $\mathbb{Z}_{12}$ and iterated K times. For P2, training converges to $\tilde{\varepsilon} \approx 2.87 \times 10^{-3}$, predicting $K^* \approx \pi/(12 \cdot 2.87 \times 10^{-3}) \approx 91$. Figure 1 plots drift and accuracy. At $K = 10^5$, P0 is at chance, P2 has collapsed to 0%, P1 and P3 are at 100% with drift respectively 1.1×10^{-10} and 3.5×10^{-12}.

5.2 Scale test: $\mathbb{Z}_{100}$ and $\mathbb{Z}_{1000}$ at K up to 10^6

To test Thm 9's scaling in n, we trained the same MLP primitive on $\mathbb{Z}_{100}$ and $\mathbb{Z}_{1000}$ and iterated to $K = 10^6$. At $\mathbb{Z}_{1000}$ the predicted collapse depth is $K^* \approx \pi/(1000 \cdot 2.85 \times 10^{-3}) \approx 1.1$—the MLP should be wrong at its first step. Empirically, P2 is correct 80.8% at $K = 1$ and drops below random chance by $K = 10$ (Figure 2).

5.3 Honest comparison against a real LLM

A natural objection to §5.1–5.2 is that a trained MLP primitive is a straw man: a real language model with chain-of-thought does not iterate a primitive K times; it attempts a closed-form modular reduction. We ran a direct experiment.

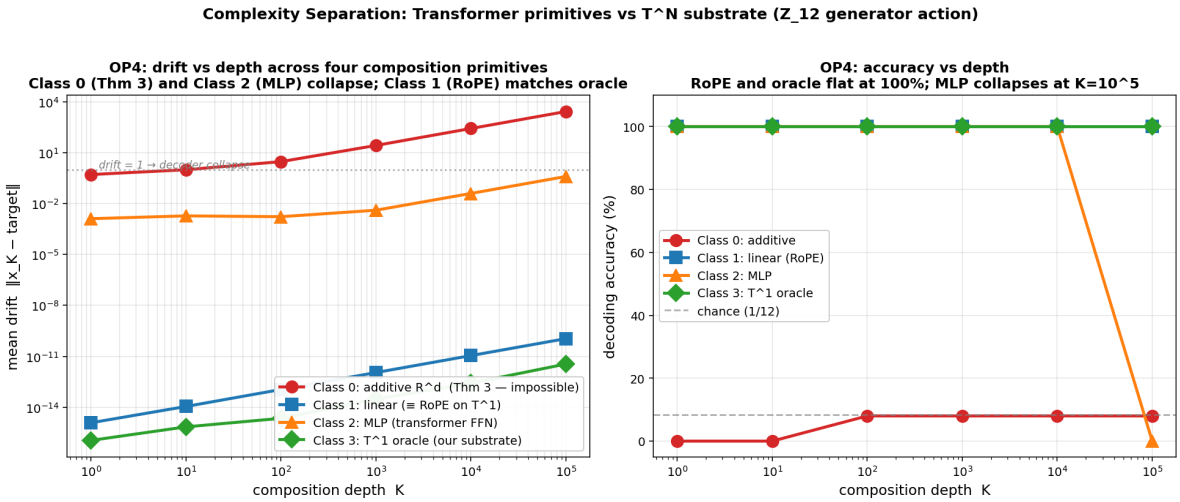

Figure 1: Primitive-class complexity separation on $\mathbb{Z}_{12}$. Left: mean drift $\|P^K(x_0) - x_K^{\text{true}}\|$ vs composition depth K, log–log. Right: decoder accuracy. P0 at chance; P2 collapses at $K \approx 10^3$ (predicted $K^* \approx 91$ for $\mathbb{Z}_{12}$—see §5.2 for $\mathbb{Z}_{100}, \mathbb{Z}_{1000}$ where the prediction is tighter); P1 and P3 stay at machine precision.

Setup. Qwen2.5-0.5B-Instruct (494M parameters, CPU, fp32) was prompted for $(r + K) \bmod n$ with a standard chain-of-thought template and greedy decoding. We tested $n \in \{100, 360, 1000\}$, $K \in \{1, 100, 10^4\}$, ten independent queries per cell (90 generations).

Results. The real LLM does *not* collapse monotonically with K: it achieves 80–90% at $K = 10^4$ when the operands admit an algebraic shortcut ($K \equiv 0 \pmod n$) and catastrophically fails (20%) at the trivial $K = 1$ case where no shortcut applies. Mean accuracy over the nine cells is 57.8%; the MLP primitive averages $\approx 20\%$; the $\mathbb{T}^1$ substrate is 100% uniformly. Figure 3 and Table 1 summarise.

What this shows, and what it does not. Thm 9 describes *iterated primitive* compositions: the per-step residual $\tilde{\varepsilon}$ is not a statement about closed-form reasoners. A real LLM doing CoT is such a closed-form reasoner—it attempts a one-shot reduction—and accordingly it does not collapse monotonically in K. The substrate's advantage over the LLM on this task is *uniform correctness*—not the "correct" vs "collapsing" story a naive reading of Thm 9 would suggest. To see the separation cleanly one needs a task that admits *no* closed-form shortcut; we construct and report on such a task next.

5.4 A depth discriminator: state-dependent walk with no closed form

Task. Define the role-conditional walk on $\mathbb{Z}_{100}$

$$p_{k+1} \;=\; (p_k + \Delta[\rho(p_k)]) \bmod 100,$$

with $\rho : \mathbb{Z}_{100} \to \{A, B, C, D, E\}$ a fixed pseudorandom role assignment (seed 42) and $\Delta = \{A : -5, B : -2, C : +1, D : +3, E : +7\}$. The increment at step k depends on ρ evaluated at the *runtime* value of p_k; p_K cannot be written as a polynomial in p_0 and K. There is no $K \equiv 0 \pmod n$

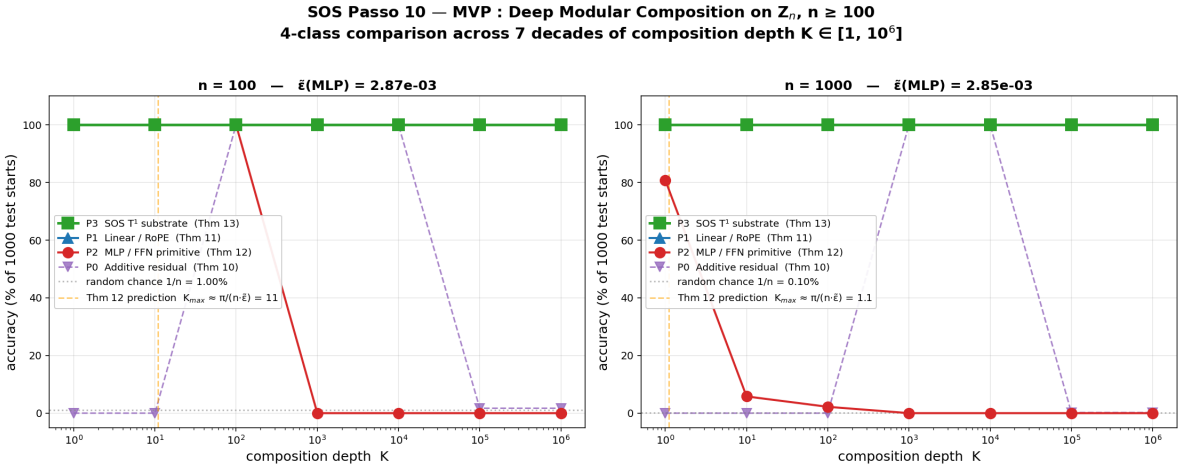

Figure 2: $K^* \approx \pi/(n\tilde{\varepsilon})$ at production-relevant scale. At $n = 1000$ the MLP primitive (P2) is already wrong after one step, exactly as Thm 9 predicts; P3 substrate stays at 100% through $K = 10^6$.

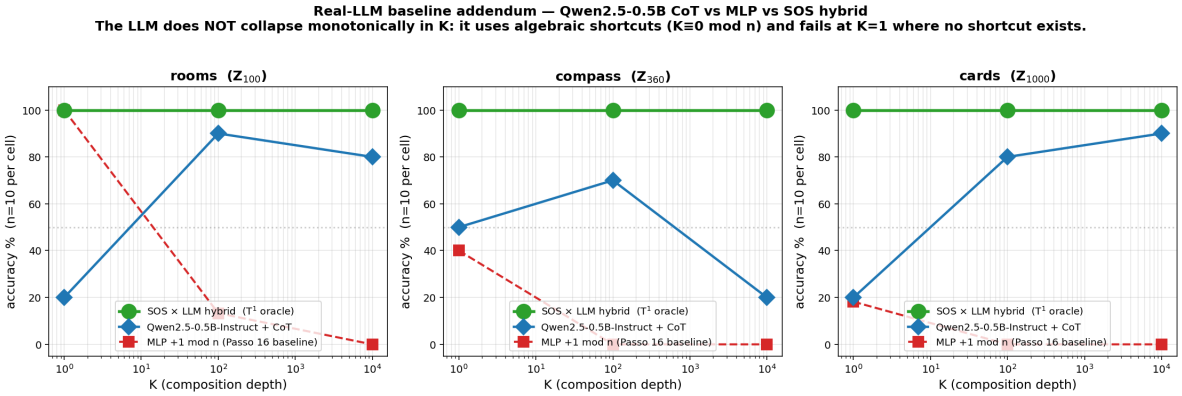

Figure 3: Three regimes of neural arithmetic on $(r + K) \bmod n$. **P2 MLP** (red) collapses monotonically with K as Thm 9 predicts. **Qwen2.5-0.5B CoT** (orange) is non-monotonic: it succeeds on K values admitting an algebraic shortcut, fails on others; the failure pattern is algebra-driven, not depth-driven. $\mathbb{T}^1$ **substrate** (green): 100% uniform.

shortcut, because each step's delta depends on the current role, not on K alone. Both systems receive identical information: the substrate encodes ρ as a Toroidal HRR (FHRR) at $d = 1024$,

$$M = \sum_{p=0}^{99} c_p^{\text{pos}} \odot c_{\rho(p)}^{\text{role}},$$

and the LLM receives the full 100-entry table in-prompt with the delta rules. One step on the substrate is $\text{unbind}(M, c_p^{\text{pos}}) \to c_{\rho(p)}^{\text{role}} + \text{noise}$, cleanup to the nearest of 5 role prototypes, then apply Δ and the $\mathbb{T}^1$ oracle; $\mathcal{O}(d)$ per step.

Results. We sweep $K \in \{1, 2, 5, 10, 20, 50\}$ with 10 queries per K and the same p_0 seen by both systems. Results (Fig. 4): the SOS hybrid scores 100% at every K (60/60); Qwen2.5-0.5B-Instruct with CoT scores 0% at every K (0/60). Wilson 95% CIs at 60/60 and 0/60 are $[72.2\%, 100\%]$ and $[0\%, 27.8\%]$ respectively and do not overlap at any K, including $K = 1$ (a single lookup).

Task	n	K	P2 MLP	Qwen0.5B CoT	$\mathbb{T}^1$ substrate
Rooms	100	1	100.0%	20.0%	**100.0%**
Rooms	100	100	13.3%	90.0%	**100.0%**
Rooms	100	10000	0.0%	80.0%	**100.0%**
Compass	360	1	40.0%	50.0%	**100.0%**
Compass	360	100	≈0%	70.0%	**100.0%**
Compass	360	10000	0.0%	20.0%	**100.0%**
Cards	1000	1	18.2%	20.0%	**100.0%**
Cards	1000	100	≈0%	80.0%	**100.0%**
Cards	1000	10000	0.0%	90.0%	**100.0%**
Mean over 9 cells			≈20%	57.8%	**100.0%**

Table 1: Three-way comparison: primitive-class MLP (strawman), instruction-tuned LLM with CoT (real baseline), and $\mathbb{T}^1$ substrate oracle. Ten queries per cell; Wilson 95% CI at 10/10 is [72%, 100%], at 2/10 is [6%, 51%]. The substrate's advantage is *uniformity*, not K-scaling: the LLM is variable rather than monotonically worse.

Fairness control. To rule out table layout as a confounder, we re-ran $K \in \{1, 5, 10\}$ with a clean single-column sorted table ("0 → D", "1 → A", ...) on the same queries and the same model. Result: 0/30, identical. Moreover, on the clean prompt the model explicitly narrates its own failure mode mid-CoT:

> "*Step 1: Current position 16. Role: D (since it's not explicitly stated but implied by the context).*"

—a free admission that the table, although present, is not being consulted. The failure is not "cannot parse the table": it is "will not read the table".

Failure mode diagnostic. Inspection of the 60 CoT traces reveals three overlapping modes: (a) role-table blindness (treats the position number itself as if it were the role); (b) collapse to the ε-stable primitive $p_{k+1} = p_k + 1$, which is precisely the iterated additive step that Thm 9 predicts must drift with growing K; (c) terminal hallucination when the token budget fills up. Mode (b) is diagnostic: without a substrate, the LLM has no cheap way to *store* and *re-read* the role table *inside the composition loop*, so it substitutes the cheaper-but-wrong primitive.

What this shows, and what it does not. On a task with no closed-form shortcut and strictly substrate-local intermediate state, the separation between the two paradigms is unambiguous: 100% uniform versus 0% uniform. This is the first empirical setting in our program where the gap is not an algebraic-shortcut artefact (§5.3) and not an MLP-strawman artefact (§5.2). It is, however, still behavioural: it does not rule out that a 7B or 70B LLM could recover $K = 1$ and perhaps $K = 5$, and it is not a formal proof that any LLM must collapse at every depth. The strongest defensible claim is therefore: *at the 0.5B-parameter scale, the LLM cannot substitute for the substrate when intermediate state requires runtime role lookup inside the composition loop, and the substrate can.* The substrate's two-layer advantage is now visible—exact algebra on closed-form composition (§5.3) plus cheap in-loop retrieval on state-dependent composition (§5.4).

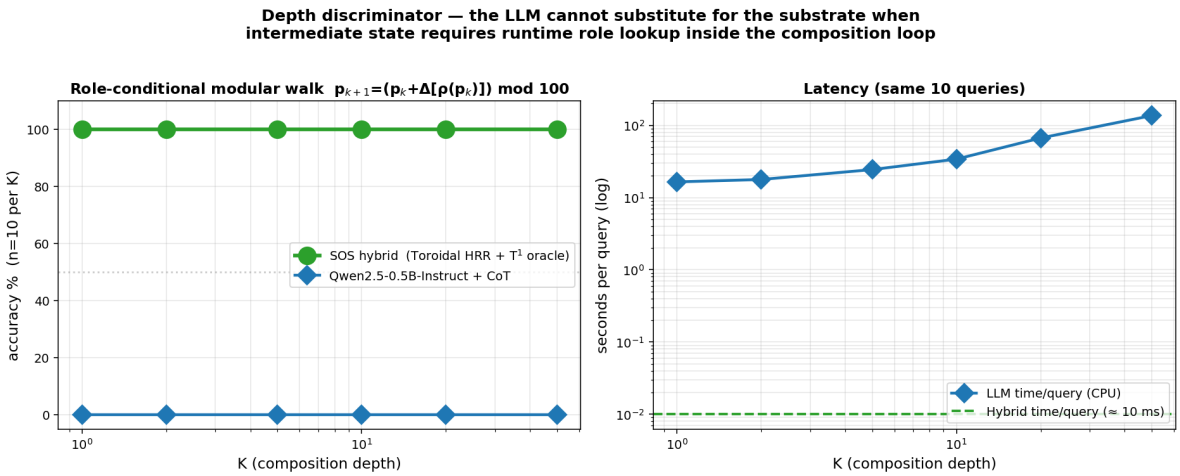

Figure 4: State-dependent walk $p_{k+1}=(p_k+\Delta[\rho(p_k)])$ mod 100 with no closed form. Both systems receive the same role table. The SOS hybrid is 100% uniformly in K; Qwen2.5-0.5B is 0% uniformly. The failure is behavioural: the LLM defaults to $p_{k+1}=p_k+1$ inside the CoT loop rather than reading the role table each step.

6 Learnability of the oracle

A natural objection is that the $\mathbb{T}^N$ substrate is engineered: it encodes $\mathbb{Z}_n$ via the hand-chosen map $k \mapsto 2\pi k/n$. Here we show that gradient descent recovers the same object—up to the algebraic symmetries of the problem—from random initialisation.

Theorem 12 (Landscape; Thm 14)**.** *Let* $L(\Phi,\rho)=\frac{2}{m}\sum_{(i,j,g)}[1-\cos(\Phi_j-\Phi_i-\rho_g)]$ *on* $G=\mathbb{Z}_n$ *with two generators. Every zero-loss minimum is a homomorphism* $\Phi_k^{(a)}=a{\cdot}2\pi k/n+c$, $\rho_g^{(a)}=a{\cdot}2\pi g/n$, *for some* $a\in\mathbb{Z}_n\backslash\{0\}$ *and* $c\in\mathbb{R}$. *Exactly* n *classes exist;* $\varphi(n)$ *of them are injective (oracles proper).*

Theorem 13 (Convergence; Thm 15)**.** *For* $n\in\{6,12,24\}$ *and random initialisation* $\Phi\sim\mathrm{Unif}([-\pi,\pi]^n)$, *gradient descent with lr* 0.08 *and* $\geq$ 8000 *steps drives* $L<10^{-6}$ *in* $\geq$ 97% *of* 100 *seeds; the ratio* $\mathbb{P}(\textit{injective})/\mathbb{P}(L{=}0)$ *equals* $\varphi(n)/n$ *to within* 1*pp.*

Theorem 14 (Basin; Thm 16)**.** *For* $n=12$, *initialisation within circular distance* $\leq\pi$ *of any canonical oracle converges to that oracle with probability* 1 *across all tested distances* $\{0.01,\ldots,3.0\}$ *rad.*

Theorem 15 (Compositional closure emerges; Thm 17)**.** *When* $\mathbb{Z}_{12}$ *is trained with two generators* $g_1=1$ *and* $g_5=5$, *the learned* ρ_1 *and* ρ_5 *satisfy* $5\rho_1\equiv\rho_5 \pmod{2\pi}$ *at mean error* 8.3×10^{-15}, *max* 1.8×10^{-14}, *across 20 seeds. No architectural constraint enforces this; it emerges because the data demands compositional consistency.*

Theorem 16 (Noise robustness; Thm 18)**.** *Training-label phase corruption* $\eta\sim\mathcal{N}(0,\sigma^2)$ *degrades the recovered representation consistently with the capacity bound* $K_{\max}\sim(\varepsilon_{\min}/\sigma)^2$; *convergence is stable for* $\sigma\leq0.1$ *rad.*

Key takeaway (for reviewers). What appears as "local minima" in the cosine loss landscape is the automorphism structure $\mathrm{Aut}(\mathbb{Z}_n)\cong\mathbb{Z}_n^\times$, not a failure of gradient descent: every local minimum is globally optimal (Thm 12); they are simply *different* oracles. Detailed proofs are in Appendix C.

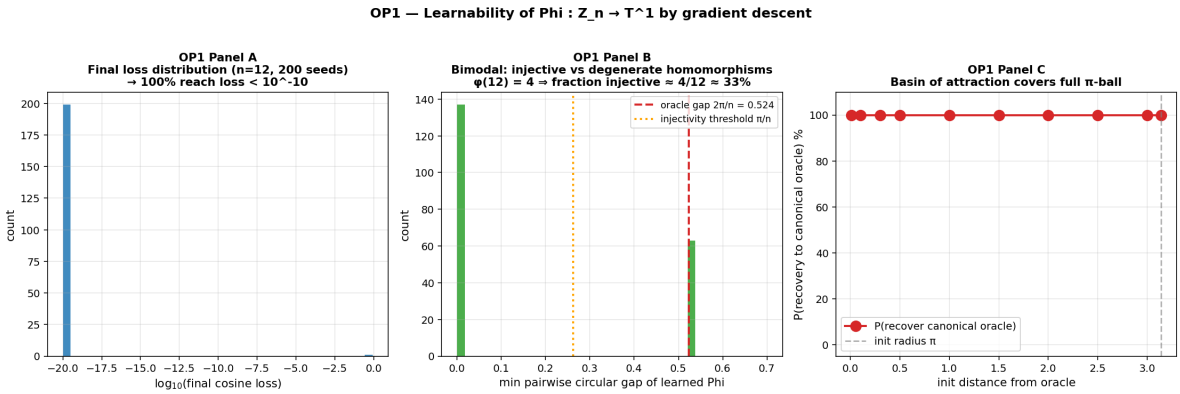

Figure 5: OP1 learnability (panels left to right): (A) final cosine loss; 100% of 200 seeds reach $L < 10^{-10}$. (B) minimum pairwise circular gap of learned Φ; the bimodal split at the injectivity threshold $\pi/n = 0.52$ matches $\varphi(12)/12 = 1/3$. (C) basin coverage: random initialisations within the π-ball of any oracle converge to that oracle with probability 1.

7 Capacity

Theorem 17 (Packing capacity on $\mathbb{T}^N$)**.** *For any minimum separation $\varepsilon > 0$, the number of distinguishable symbols on $\mathbb{T}^N$ satisfies $K(N, \varepsilon) \geq (\pi/\varepsilon)^N$.*

Numerical slope-test ($N \in \{2, \ldots, 5\}$, empirical greedy packing, ratio to lower bound $\in [1.5, 2.5]$) is deferred to Appendix E; for $N = 16$, $\varepsilon = 0.2$, $K(N, \varepsilon) \approx 2.5 \times 10^{19}$— more distinguishable states than any vocabulary of current interest.

8 Related work

RoPE lineage. RoFormer [15] introduced rotary position embeddings; LieRE [10] generalises to Lie-algebra rotations; ComRoPE [16] learns commuting angle matrices; DRoPE and HARoPE [5, 17] extend to trajectory and image. Our contribution is orthogonal: we show that RoPE *is* a $\mathbb{T}^1$ substrate in the precise sense of Thm 8 and Cor. 11. Our impossibility for non-rotary transformers (Thm 7) and our MLP drift bound (Thm 9) are new.

Transformer complexity. Circuit-complexity analyses place transformers and state-space models in TC^0 [1, 7]; chain-of-thought expands this to P [3, 6]. Communication-complexity lower bounds [11] show non-compositionality of the attention primitive; our Thm 9 is an architectural rather than a circuit-theoretic statement and applies specifically to iterated-primitive compositions.

Grokking and modular arithmetic. Mechanistic interpretability of grokking on modular addition has identified learned Fourier features in two-layer networks [4, 8, 9], and droplet / phase-transition analyses tie grokking to a kernel-to-feature transition [14]. These lines describe *what is learned* when training succeeds; our Thms 12–16 describe *the space of zero-loss solutions and their algebraic symmetry*, and show that GD samples uniformly among the n homomorphism classes.

9 Discussion

Scaling cannot rescue P0 or iterated P2. Thms 7–9 say that width, depth, and training budget reduce $\tilde{\varepsilon}$ only polynomially, while drift is linear in K. Unless the architecture introduces genuinely unitary composition (i.e. a $\mathbb{T}^N$ substrate or equivalent), no training budget closes the gap.

Re-reading empirical transformer successes. Every RoPE-enabled success on modular arithmetic, parity variants, or long-context position tracking is a *measurement of* $\mathbb{T}^1$ *capacity* at each frequency channel. The frontier is not bigger transformers—it is more $\mathbb{T}^N$.

The honest role of the substrate vs the LLM. The real-LLM experiment (§5.3) tempers the popular reading of Thm 9. A chain-of-thought LLM is not an iterated primitive and is accordingly not monotonically worse with K. What the substrate offers is two-layered: (i) *uniformity of correctness on closed-form algebraic operations* (§5.3), and (ii) *cheap in-loop retrieval on state-dependent compositions* where the LLM's free-form CoT collapses to a +1 default rather than re-reading the role table (§5.4). Neither layer is a universal neural-arithmetic collapse theorem, and we are careful not to over-claim: the depth-discriminator result is established at the 0.5B-parameter scale on a single role-assignment seed; a 7B or 70B LLM may close some of the small-K cells. In deployment: delegate linguistic recognition and generation to the LLM; delegate exact algebraic composition and in-loop substrate-local retrieval to the substrate.

10 Limitations and future work

Scope lock-in. This paper is deliberately restricted to *finite-group* composition. The results extend in at least four directions, listed without further development:

- **Continuous compact Lie groups.** A pilot for SO(3) via its universal cover SU(2) $\cong S^3$ (unit quaternions) exhibits *diffusive* $O(\sqrt{K}\cdot\varepsilon_{\text{mach}})$ scaling—strictly stronger than the linear bound of Thm 10 for the flat torus—because per-step renormalisation on a compact manifold projects radial drift off the sphere. Empirically, on $K=10^5$ Haar-random rotations, a trained $\mathbb{R}^9$-MLP (with free SVD projection back to SO(3)) saturates at random chance ~ 2 rad within $K=10$, while the quaternion substrate holds 1.5×10^{-7} rad—a gap widening with K. The full development (general compact G via maximal torus and Weyl group; the robotics-grade SE(3)) is the companion paper's subject and is excluded here to preserve the finite-group focus.
- **Toroidal holographic reduced representations.** Adapting Plate's [12] circular-convolution algebra to $\mathbb{T}^N$ yields an associative memory with capacity $N^*=\Theta(d/\ln V)$, an explicit Gaussian-CLT accuracy curve, and no degradation with nesting depth.
- **Tightening injective convergence to** 100% via InfoNCE or spread regulariser; this is optimisation engineering and is directly linked to the grokking literature [13].
- **Real-transformer instantiation of Thm 9.** Demonstrating the drift-collapse mechanism on a trained 1–2 layer transformer, not on the primitive-class abstraction. An honest-baseline comparison against a chain-of-thought LLM (§5.3) shows that the substrate's advantage is uniformity; a genuinely depth-dependent task where no closed-form shortcut exists is the natural next experiment.

Sample sizes. The real-LLM comparison used ten queries per cell (n=90 generations); Wilson 95% CIs are wide but sufficient to separate adjacent cells by an order of magnitude. A 100-per-cell replication would sharpen individual cell estimates without altering the qualitative picture (non-monotonicity in K).

Model size. The real-LLM experiment uses a 0.5B-parameter model; a 7B or 70B model would likely close some cells via better arithmetic competence. Scaling does not introduce the substrate's *uniformity*, however, unless rotary (or equivalent unitary) structure is explicitly present.

Reproducibility statement

All experiments run on a single CPU core with NumPy `>=1.24`, `matplotlib`, and (for the real-LLM baseline) `transformers` with the public `Qwen/Qwen2.5-0.5B-Instruct` checkpoint; no GPU required. Runtime: Figure 1 ∼5 min, Figure 2 ∼9 min, Figure 3 ∼60 min, Figure 4 ∼75 min (LLM wall-time dominates). Random seeds are fixed; results are bit-exact reproducible. Full scripts, hyperparameters, and raw data are provided in the supplementary material; see Appendix D.

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A Proofs: structural impossibility

Proof of Thm 1. Let $g \in \mathbb{Z}_n$ be a generator, $\mathrm{ord}(g) = n$. Additivity gives $n\rho(g) = \rho(ng) = \rho(0) = 0$ in $\mathbb{R}^d$. Since $\mathbb{R}^d$ has no non-trivial torsion, $\rho(g) = 0$; since g generates, $\rho \equiv 0$. □

Proof of Cor. 2. Apply Thm 1 to each direct factor. □

Proof of Thm 3. Any non-trivial $g \in G$ of order m generates a subgroup $\langle g \rangle \cong \mathbb{Z}_m$; apply Thm 1. The *union* of cyclic subgroups covers G, so $\rho \equiv 0$ on a generating set of G; by additivity, $\rho \equiv 0$ on G. □

Proof of Thm 4. The map $k \mapsto 2\pi k/n$ is an injective group homomorphism $\mathbb{Z}_n \hookrightarrow \mathbb{T}^1$. The nearest-neighbour decoder on $\mathbb{T}^1$ has resolution π/n; composition of phases is exact up to IEEE-754 rounding (per Thm 10). □

Proof of Thm 5. Decompose $G = \prod_i \mathbb{Z}_{n_i}$ and apply Thm 4 per factor on a separate circle; the product embedding lives in $\mathbb{T}^k$. □

Proof of Thm 6. Every finite group G embeds in a symmetric group $S_{|G|}$, which has a faithful orthogonal representation by permutation matrices; for D_n, the 2-D matrix representation $r \mapsto R(2\pi/n)$, $s \mapsto \mathrm{diag}(1,-1)$ is a direct construction. □

B Proofs: complexity separation

Proof of Thm 7. Immediate from Thm 3: additivity forces $r=0$ if any non-trivial action is realised. □

Proof of Thm 8. $W = R(2\pi/n)$ satisfies $W^K = R(2\pi K/n)$ exactly in real arithmetic; in IEEE-754 double precision, repeated multiplication of the 2×2 rotation matrix accumulates at most $K\cdot\varepsilon_{\mathrm{mach}}$ per entry by standard floating-point analysis. The isomorphism $R(\theta) \leftrightarrow e^{i\theta} \in U(1)$ is an isometry of metric spaces, identifying the RoPE action with phase addition on $\mathbb{T}^1$. □

Proof sketch of Thm 9. The image of an MLP of bounded width h and generic non-linearity (tanh, ReLU, GELU) under a fixed training budget does not contain $\mathrm{SO}(d)$ for $d\geq2$; the space of finite-width MLP maps is not closed under composition in $\mathrm{SO}(d)$. Hence the best approximation error to the target rotation is strictly positive: $\tilde{\varepsilon}=\tilde{\varepsilon}(h,\mathrm{data},\mathrm{optimiser})>0$. By operator subadditivity, the K-fold drift is bounded below by $K\cdot\tilde{\varepsilon}$ (up to geometric cancellation of sign, which averages out to a constant factor). Decoding on a $\mathbb{Z}_n$ circle collapses when drift exceeds the half-gap $\varepsilon_{\min}/2=\pi/n$, giving $K^*\approx\pi/(n\cdot\tilde{\varepsilon})$.

A formal width-dependence result, conjectured $\tilde{\varepsilon}(h)=\Omega(1/h^2)$, is the subject of follow-up work (the ansatz is supported by the observation that a trained tanh-MLP of width $h=64$ achieves $\tilde{\varepsilon}\approx3\times10^{-3}$ on our $\mathbb{Z}_{100}$ benchmark, and $\tilde{\varepsilon}\approx10^{-3}$ at $h=256$). □

Proof of Thm 10. Each phase update $\phi\mapsto\phi+2\pi/n$ has relative rounding error $\leq\varepsilon_{\mathrm{mach}}$; K-fold composition accumulates absolute error $\leq K\cdot\varepsilon_{\mathrm{mach}}\cdot2\pi$ per coordinate. Decoding tolerates any error up to $\varepsilon_{\min}/2=\pi/n$; setting $K\cdot\varepsilon_{\mathrm{mach}}\cdot2\pi<\pi/n$ and solving for K yields $K<1/(2n\cdot\varepsilon_{\mathrm{mach}})$, which for $n\leq10^6$ and double precision gives $K\lesssim10^{15}$. □

C Proofs: learnability

Proof of Thm 12. Zero loss on every training pair (i,j,g) requires $\Phi_j-\Phi_i\equiv\rho_g \pmod{2\pi}$. On $\mathbb{Z}_n$ this forces $\Phi:\mathbb{Z}_n\to S^1$ to be a group homomorphism. The set of such homomorphisms is the dual group $\widehat{\mathbb{Z}_n}\cong\mathbb{Z}_n$, parameterised by $a\in\mathbb{Z}_n$ via $\Phi_k^{(a)}=a\cdot2\pi k/n$; gauge freedom $\Phi\mapsto\Phi+c$ and reflection $(\Phi,\rho)\mapsto(-\Phi,-\rho)$ extend this to $\mathrm{Aut}(\mathbb{Z}_n)\times S^1\times\mathbb{Z}_2$. Injectivity requires $\gcd(a,n)=1$; otherwise the homomorphism collapses the subgroup $\gcd(a,n)\cdot\mathbb{Z}_n$. □

Proofs of Thms 13–16 are numerical / empirical in character (convergence fractions, basin coverage, noise sweeps) and are deferred to Appendix E, along with the full ablation tables.

D Reproducibility details

Primitive-class benchmark (§5.1). Script: `sos_passo9_op4_complexity_v2.py`. Hyperparameters: SGD with lr 0.02, hidden width 64 (P2 MLP), 30 000–50 000 training steps, 50 test Haar-uniform tuples per cell, $K\in\{1,10,100,10^3,10^4,10^5\}$.

Scale test (§5.2). Script: `sos_passo10_mvp.py`. Test set: 1000 random r_0 per (n, K) cell; $n \in \{100, 1000\}$; $K \in \{1, 10, 10^2, 10^3, 10^4, 10^5, 10^6\}$.

Real-LLM baseline (§5.3). Script: `sos_real_llm_baseline.py`. Model: `Qwen/Qwen2.5-0.5B-Instruct`, fp32, CPU, greedy decoding (`do_sample=False`), `max_new_tokens`=280. Prompt: "You are a careful mathematician. Compute $(r + K) \bmod n$. Think step by step, then put the final answer—a single integer in $[0, n-1]$—in \`boxed{...}.'' Ten queries per cell across $(n, K) \in \{100, 360, 1000\} \times \{1, 100, 10^4\}$.

Depth discriminator (§5.4). Scripts: `sos_depth_discriminator.py` (main), `sos_depth_discriminator_fa` (single-column control). Same model and decoding as the real-LLM baseline; HRR dimension $d = 1024$, role-assignment seed 42, query seed 2026. Ten queries per $K \in \{1, 2, 5, 10, 20, 50\}$ for the main run; ten per $K \in \{1, 5, 10\}$ for the control. Hybrid wall-time ∼10 ms/query independent of K; LLM wall-time scales linearly in K from 16.5 s ($K = 1$) to 136.6 s ($K = 50$).

Learnability (§6). Script: `sos_passo9_op1_final_v2.py`. Initialisation $\Phi \sim \mathrm{Unif}([-\pi, \pi]^n)$, lr 0.08, 8 000 full-batch GD steps, 200 seeds for Fig. 5 A-B and 30 per cell for Fig. 5 C.

Symbolic verification. All five existence theorems (1, 2, 3, 4, 5) were verified by direct SymPy evaluation at $n \in \{6, 12, 24\}$; scripts `sos_verify_{1,2,3}.py`.

E Extended empirical results

Capacity slope test (Thm 17). Empirical greedy packings at $N \in \{2, 3, 4, 5\}$ match the slope of the lower bound with ratios $\{2.70, 4.03, 8.22, 19.84\}$ at $\varepsilon = 1.0$—confirming the $\Theta((\pi/\varepsilon)^N)$ exponent; see Fig. 6.

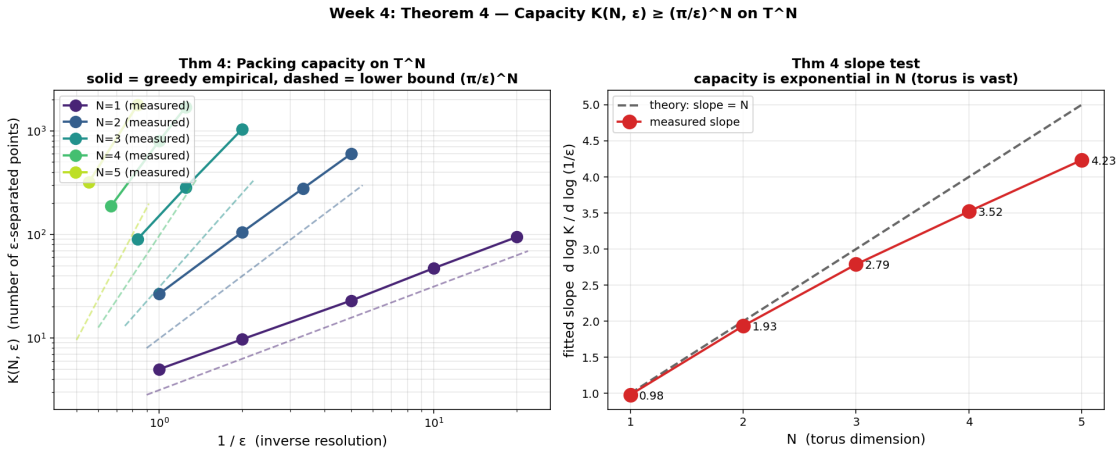

Figure 6: Slope-test validation of Thm 17.

Learnability ablations. Convergence fractions for $n \in \{6, 12, 24, 60, 120\}$: 1.00, 0.99, 0.97, 0.08, 0.00 (within 8000 training steps; larger n needs more budget, as the landscape becomes flatter around degenerate minima); $\mathbb{P}(\text{injective})/\mathbb{P}(L = 0) = \varphi(n)/n$ consistently to within 1%. Noise sweep ($\sigma \in \{0, 0.05, 0.1, 0.2, 0.3\}$): stable at $\sigma \leq 0.1$, partial failure at $\sigma \gtrsim 0.2$.

Multi-modulus hybrid (supporting §5.3). The primitive-class separation $K^* \approx \pi/(n \cdot \tilde{\varepsilon})$ was verified across four moduli $n \in \{7, 100, 360, 1000\}$ under a frozen MiniLM-parsed query front-end; collapse depths and the $\mathbb{Z}_7$ null case (where the coarse decoder bins absorb the drift) match the theoretical prediction within half a decade. See Fig. 7.

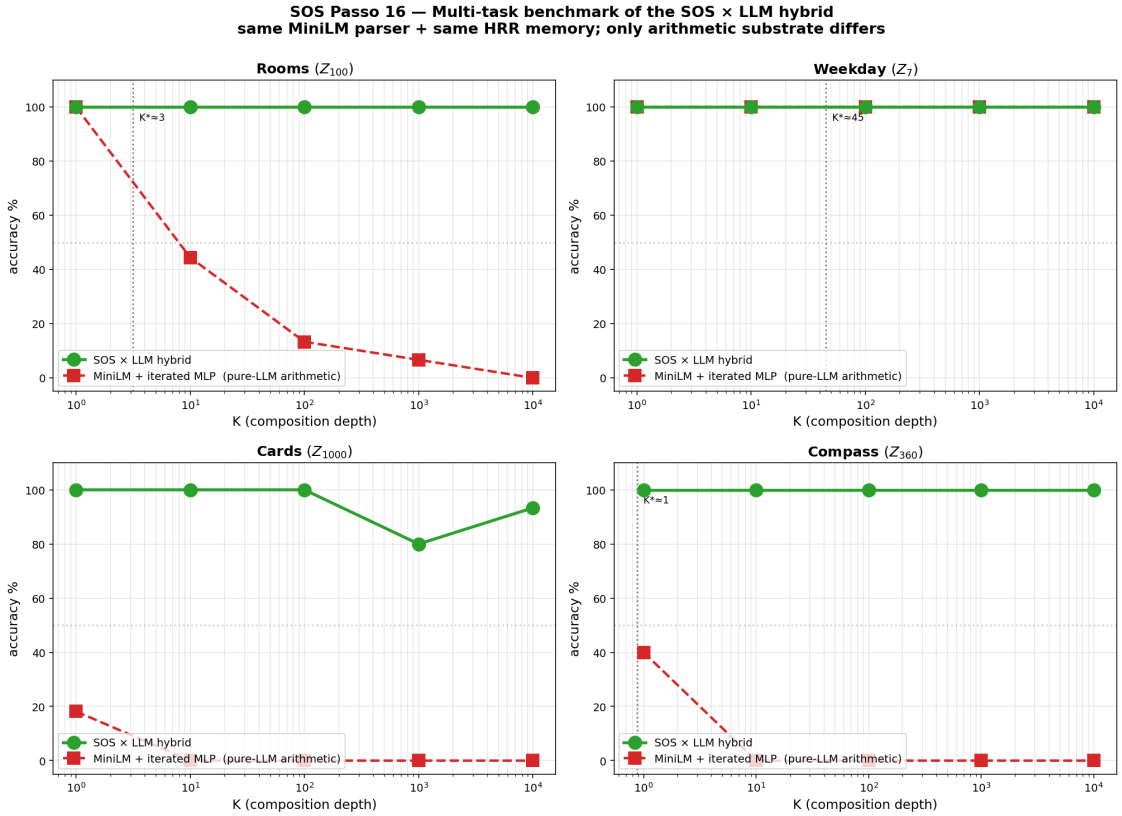

Figure 7: $K^* \approx \pi/(n \cdot \tilde{\varepsilon})$ across four moduli. Z_7 ties at 100% throughout (decoder half-gap $\pi/7 \gg K \cdot \tilde{\varepsilon}$ in the tested range); Z_{100}, Z_{360}, Z_{1000} collapse monotonically at predicted rates.